Mirpur University of Science and Technology

Department of Software Engineering

Project Report

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Section: A

Session: 23-27

Subject: Web Engineering

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UniConnect – Campus Event Management System

1. Project Title & Introduction

Project Title:

UniConnect – Campus Event Management System

Introduction:

UniConnect is a full-stack web application designed to manage campus-related activities such as events, announcements, and shared resources in a centralized system.

The system provides an easy-to-use interface for administrators to add, view, and manage campus information, while ensuring data consistency and accessibility.

The project aims to reduce manual effort, improve communication within campus communities, and demonstrate practical implementation of full-stack web development concepts.

2. Problem Statement

In many educational institutions, campus events and announcements are managed manually through notice boards, WhatsApp groups, or scattered documents. This leads to:

- Lack of centralized information
- Missed announcements and events
- Difficulty in managing and updating data
- No proper digital record

Problem:
There is a need for a centralized, web-based system to manage campus events, announcements, and shared resources efficiently.

3. System Features

The UniConnect system provides the following features:

Event Management

- Add new campus events
- View all events in a dashboard
- Delete events
- Event date tracking

Announcement Management

- Add announcements
- View all announcements
- Delete outdated announcements

Resource Management

- Add useful resources (Drive links, PDFs, URLs)
- View available resources
- Delete resources

Dashboard

- Displays total number of:
    - Events
    - Announcements
    - Resources
- Clean and professional UI

General Features

- Responsive UI
- CRUD operations
- RESTful API structure
- GitHub version control

4. Technology Stack

Frontend

- HTML5
- CSS3 (Custom UI/UX styling)
- JavaScript (Vanilla JS)

Backend

- Node.js
- Express.js

Database

- MySQL

Tools & Platforms

- Git & GitHub
- Visual Studio Code
- MySQL Workbench
- Browser (Chrome)

5. Frontend Design Explanation

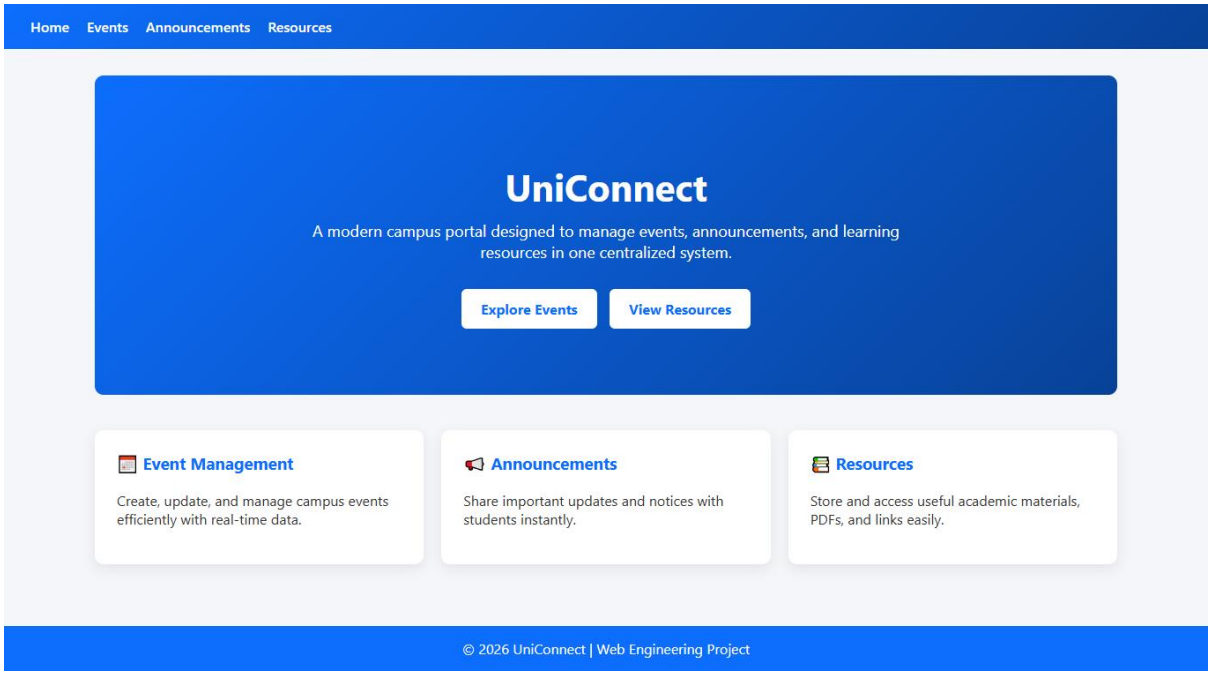
The frontend is built using **HTML, CSS, and JavaScript**.

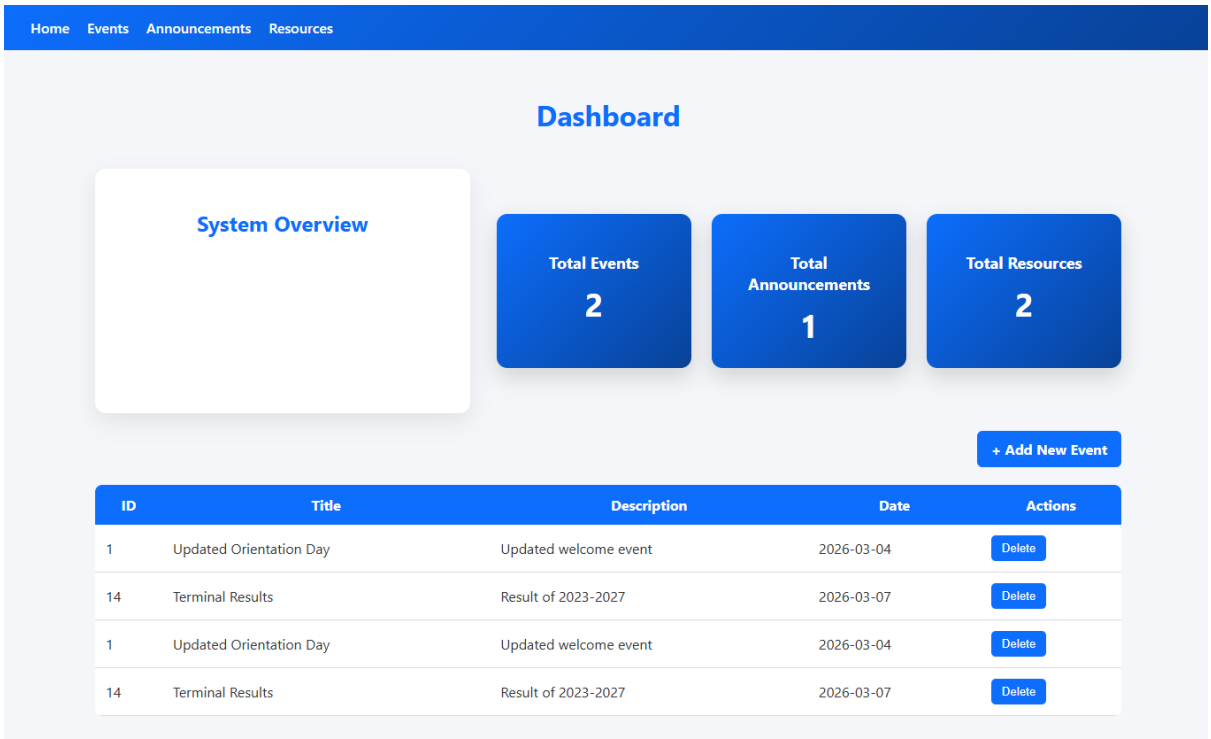
Key Pages

1. Home Page (
2. Dashboard Page
3. Add Event Page
4. Announcements Page
5. Resources Page

Design Highlights

- Gradient navigation bar
- Centered animated forms
- Card-based dashboard layout
- Hover effects and transitions
- Clean and professional color theme


Home Events Announcements Resources
UniConnect
A modern campus portal designed to manage events, announcements, and learning resources in one centralized system.
Explore Events
View Resources
Event Management
Create, update, and manage campus events efficiently with real-time data.
Announcements
Share important updates and notices with students instantly.
Resources
Store and access useful academic materials, PDFs, and links easily.
© 2026 UniConnect | Web Engineering Project


Home Events Announcements Resources
Dashboard
System Overview
Total Events
2
Total Announcements
1
Total Resources
2
+ Add New Event
ID Title Description Date Actions
1 Updated Orientation Day Updated welcome event 2026-03-04 Delete
14 Terminal Results Result of 2023-2027 2026-03-07 Delete
1 Updated Orientation Day Updated welcome event 2026-03-04 Delete
14 Terminal Results Result of 2023-2027 2026-03-07 Delete

Home Events Announcements Resources

Announcements

+ Add Announcement

ID	Title	Message	Date	Actions
2	Eidulfitr	Eidulfitar was Coming	2026-02-28	Delete
2	Eidulfitr	Eidulfitar was Coming	2026-02-28	Delete

Home Events Announcements Resources

Add New Event

Create and publish a new campus event

Event Title

e.g. Orientation Day

Description

Event details...

Event Date

mm/dd/yyyy

Add Event

Home Events Announcements Resources

Add Announcement

Title

Message

Add Announcement

Home Events Announcements Resources

Resources

+ Add Resource

ID	Title	Type	Link	Date	Actions
3	Github	PDF	Open	2026-03-01	Delete
2	Elementor Pro	PDF	Open	2026-02-28	Delete
3	Github	PDF	Open	2026-03-01	Delete
2	Elementor Pro	PDF	Open	2026-02-28	Delete

Home Events Announcements Resources

Add Resource

Upload or link a useful academic resource

Title

e.g. Data Structures Notes

Type

PDF / Video / Link

Resource Link

https://example.com

Add Resource

6. Backend Architecture

The backend follows a **RESTful architecture** using Express.js.

Backend Structure:

- server.js – Main server file
- routes/ – API route handlers
  - events routes
  - announcements routes
  - resources routes
- Database connection using MySQL

API Endpoints

- GET /api/events
- POST /api/events/add
- DELETE /api/events/delete/:id
- Similar routes for announcements and resources

7. Database Design

The database is designed using **MySQL**.

Tables

1. events
   - id
   - title
   - description
   - event_date
   - created_at

2. announcements
   - id
   - title
   - message
   - created_at

3. resources
   - id
   - title
   - type
   - link
   - created_at

Relationships:

- Independent tables
- Each table supports full CRUD operations

8.

```
mysql> USE uniconnect;
Database changed
mysql> SHOW TABLES;
+-----------------------+
| Tables_in_uniconnect |
+-----------------------+
| announcements         |
| events                |
| resources             |
+-----------------------+
3 rows in set (0.04 sec)

mysql> _
```

```
+-----------------------+
3 rows in set (0.04 sec)

mysql> DESCRIBE events;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| id          | int          | NO   | PRI | NULL    | auto_increment |
| title       | varchar(200) | YES  |     | NULL    |                |
| description | text         | YES  |     | NULL    |                |
| event_date  | date         | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
4 rows in set (0.04 sec)

mysql>
```

```
4 rows in set (0.04 sec)

mysql> DESCRIBE announcements;
+------------+--------------+------+-----+-------------------+-------------------+
| Field      | Type         | Null | Key | Default           | Extra             |
+------------+--------------+------+-----+-------------------+-------------------+
| id         | int          | NO   | PRI | NULL              | auto_increment    |
| title      | varchar(255) | NO   |     | NULL              |                   |
| message    | text         | NO   |     | NULL              |                   |
| created_at | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+--------------+------+-----+-------------------+-------------------+
4 rows in set (0.01 sec)

mysql> _
```

```
mysql> DESCRIBE resources;
+------------+--------------+------+-----+-------------------+-------------------+
| Field      | Type         | Null | Key | Default           | Extra             |
+------------+--------------+------+-----+-------------------+-------------------+
| id         | int          | NO   | PRI | NULL              | auto_increment    |
| title      | varchar(255) | NO   |     | NULL              |                   |
| type       | varchar(100) | YES  |     | NULL              |                   |
| link       | text         | NO   |     | NULL              |                   |
| created_at | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)

mysql>
```

```
5 rows in set (0.00 sec)

mysql> SELECT * FROM events;
+----+-------------------------+-----------------------+------------+
| id | title                   | description           | event_date |
+----+-------------------------+-----------------------+------------+
|  1 | Updated Orientation Day | Updated welcome event | 2026-03-05 |
| 14 | Terminal Results        | Result of 2023-2027   | 2026-03-08 |
+----+-------------------------+-----------------------+------------+
2 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM announcements;
+----+------------+-----------------------+---------------------+
| id | title      | message               | created_at          |
+----+------------+-----------------------+---------------------+
|  2 | Eidulfitr  | Eidulfitar was Coming | 2026-02-28 11:50:27 |
+----+------------+-----------------------+---------------------+
1 row in set (0.01 sec)

mysql>
```

```
mysql> SELECT * FROM resources;
+----+---------------+------+--------------------------------+---------------------+
| id | title         | type | link                           | created_at          |
+----+---------------+------+--------------------------------+---------------------+
|  2 | Elementor Pro | PDF  | https://example.com/web-notes.pdf | 2026-02-28 16:07:01 |
|  3 | Github        | PDF  | https://github.com/web-notes.pdf  | 2026-03-01 12:07:22 |
+----+---------------+------+--------------------------------+---------------------+
2 rows in set (0.01 sec)

mysql>
```

```
mysql> SELECT * FROM events ORDER BY event_date DESC;
+----+-------------------------+-----------------------+------------+
| id | title                   | description           | event_date |
+----+-------------------------+-----------------------+------------+
| 14 | Terminal Results        | Result of 2023-2027   | 2026-03-08 |
|  1 | Updated Orientation Day | Updated welcome event | 2026-03-05 |
+----+-------------------------+-----------------------+------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM announcements ORDER BY created_at DESC;
+----+------------+-----------------------+---------------------+
| id | title      | message               | created_at          |
+----+------------+-----------------------+---------------------+
|  2 | Eidulfitr  | Eidulfitar was Coming | 2026-02-28 11:50:27 |
+----+------------+-----------------------+---------------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM resources ORDER BY created_at DESC;
+----+---------------+------+---------------------------------+---------------------+
| id | title         | type | link                            | created_at          |
+----+---------------+------+---------------------------------+---------------------+
|  3 | Github        | PDF  | https://github.com/web-notes.pdf  | 2026-03-01 12:07:22 |
|  2 | Elementor Pro | PDF  | https://example.com/web-notes.pdf | 2026-02-28 16:07:01 |
+----+---------------+------+---------------------------------+---------------------+
2 rows in set (0.00 sec)

mysql>
```

GitHub Project Repository

The complete project is managed using GitHub for version control

https://github.com/arifmuneeb81-svg/UniConnect-Campus-Event-Management-System

GitHub Usage

- Initialized Git repository
- Commits for feature development
- Remote push to GitHub
- Public repository for evaluation

9. Conclusion & Future Enhancements

Conclusion:

UniConnect successfully demonstrates a full-stack web application using modern web technologies. The system efficiently manages campus events, announcements, and resources with a clean user interface and reliable backend architecture.
The project fulfills all functional requirements and provides a strong foundation for real-world campus management systems.

Future Enhancements:

- User authentication (Admin/Login)
- Role-based access
- Event registration for students
- Email or SMS notifications
- Deployment on cloud (Render / Railway / AWS)
- Search and filter options
- Mobile app integration